

BCS Phonon Resonance at ω_{SCm}

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PAPER_955: BCS Phonon Resonance at ω_{SCm}

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Session: 214

Source: et_phonon_resonance.py §6 (BCSPhononResonance)

Calculator: BCSPhononResonanceCalc (CP4 #539)

CVW: v2.0.0 compliant

Abstract

We analyze the BCS superconducting gap at the SCm phonon resonance frequency $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz. The gap equation $\Delta = (\hbar\omega_{\text{SCm}}/2) \tanh(\Delta / 2k_{\text{BT}}) \cdot S_{\{26\}} \cdot (F_{\text{UBi}}/F_{\text{U}})$ is solved self-consistently at the phonon resonance, revealing how the SCm vacuum phonon mode mediates Cooper pairing.

1. BCS Gap at Phonon Resonance

$$\Delta_{\text{res}} = \frac{\hbar\omega_{\text{SCm}}}{2} \cdot \tanh\left(\frac{\Delta}{2k_{\text{BT}}}\right) \cdot S_{\{26\}} \cdot \frac{F_{\text{UBi}}}{F_{\text{U}}}$$

2. Iterative Solution

Starting from $\Delta_0 = \hbar\omega_{\text{SCm}}/2$, iterate:

$$\Delta_{n+1} = \text{rhs}(\Delta_n, T)$$

until $|\Delta_{n+1} - \Delta_n| < \epsilon$.

3. Phonon Enhancement Factor

The resonance Q-factor at the phonon peak:

$$Q_{\text{res}} = \frac{\omega_{\text{SCm}}}{\sqrt{\Delta}} \cdot k_{\text{BT}}$$

4. Source Data

- **File:** et_phonon_resonance.py §6

- **Session:** 214

- **CP4 Class:** BCSPhononResonanceCalc (#539)

References

1. Bardeen, Cooper, Schrieffer — Theory of Superconductivity (1957)
2. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
3. PAPER_949 — BCS Gap Equation
4. PAPER_950 — BCS Critical Temperature
5. PAPER_951 — Cooper Pair Phonon Coupling
6. PAPER_956 — Spectral Ladder Phonon Mapping

Cross-Links

Related Paper Relationship
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PAPER_949 Gap equation solved at resonance
PAPER_951 Coupling strength at phonon peak
PAPER_956 26-level phonon mapping uses this Q
PAPER_952 Spectral ladder energies

Session 225: Late-Corpus Physics Integration (PAPER_1000-1081)

> The following physics upgrades incorporate equations, mechanisms, and
> derivations from the late-corpus papers (Sessions 219-225, PAPER_1000-1081).
> These represent body-level integrations of phonon physics, buoyancy
> formulations, and S26(3) Ramanujan corrections into this paper's domain.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and
> PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific
equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$
(matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector Domain Late-Corpus Result
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1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	$F_{U_{Bi}}$ buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)
 <!-- PKG-S26-S225 -->

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and
 > PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078
 > (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{SSq}{e^{-\kappa_{i,n/26}}} \right]$$

where $(a)_n = a(a+1) \cdots s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \ll W_{26}(n) = \prod_{i=1}^{26} \left[1 + \frac{SSq}{e^{-\kappa_{i,n/26}}} \right]$$

This sum converges absolutely for $|SSq| < 1$ (satisfied by $|SSq| = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $SSq \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa_{i,n/26}}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
SSq	—	0.57	BH dynamics
ω_{SCm}	—	$2\pi \times 1.25 \text{ THz}$	Resonance frequency
F_{UBi}/F_U	—	0.6	Buoyancy ratio

SM Anchor — CVW v2.0.0

Observable UQFF Prediction Status
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Δ_{res} at ω_{SCm} Self-consistent via fixed-point Validated
Phonon Q-factor $Q = \omega_{\text{SCm}} \sqrt{\Delta} / (k_{\text{BT}})$ Derived

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification ***Sector:** Phonon Resonance (BCS at ω_{SCm})*

§A.2 Lagrangian Density $\mathcal{L}_{\text{res}} = \mathcal{L}_{\text{gap}} \big|_{\omega=\omega_{\text{SCm}}} + \frac{1}{2} \hbar \omega_{\text{SCm}} \coth \left(\frac{\hbar \omega_{\text{SCm}}}{2k_{\text{BT}}} \right)$

§A.3 Euler-Lagrange Equation of Motion $\boxed{\Delta_{\text{res}}} = \frac{\hbar \omega_{\text{SCm}}}{2} \tanh \left(\frac{\Delta}{2k_{\text{BT}}} \right) S_{\{26\}} \frac{F_{\text{UBi}}}{F_{\text{U}}}$

§A.4 Cosmogenesis Linkage Chain *PAPER_877 $\rightarrow$ SCm vacuum $\rightarrow$ phonon peak $\rightarrow$ BCS gap at resonance $\rightarrow$ maximal Cooper pairing*

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS On-resonance VDS peak: $\rho_{\text{SCm}} \cdot S_{\{26\}}$ (maximal phonon occupation).

§B.2 DVP Resonance prime $p = 2$ (Cooper pair at peak coupling).

§B.3 BSH Q_{res} determines the BSH bandwidth at the resonance shell.

§B.4 Production-Scale Consistency

Metric Value Status
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ω_{SCm} $2\pi \times 1.25$ THz Confirmed
$[SSq]$ 0.57 Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1003	Spectral Ladder Merger 26-State Hierarchy
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay

12 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Ashcroft, N.W. & Mermin, N.D. (1976). *Solid State Physics*. Harcourt
4. Kittel, C. (2004). *Introduction to Solid State Physics*. 8th ed. Wiley
5. Feynman, R.P. (1982). *Simulating Physics with Computers*. Int. J. Theor. Phys. **21**, 467 — doi:10.1007/BF02650179
6. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem*. Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
7. Weinberg, S. (1989). *The Cosmological Constant Problem*. Rev. Mod. Phys. **61**, 1 — doi:10.1103/RevModPhys.61.1